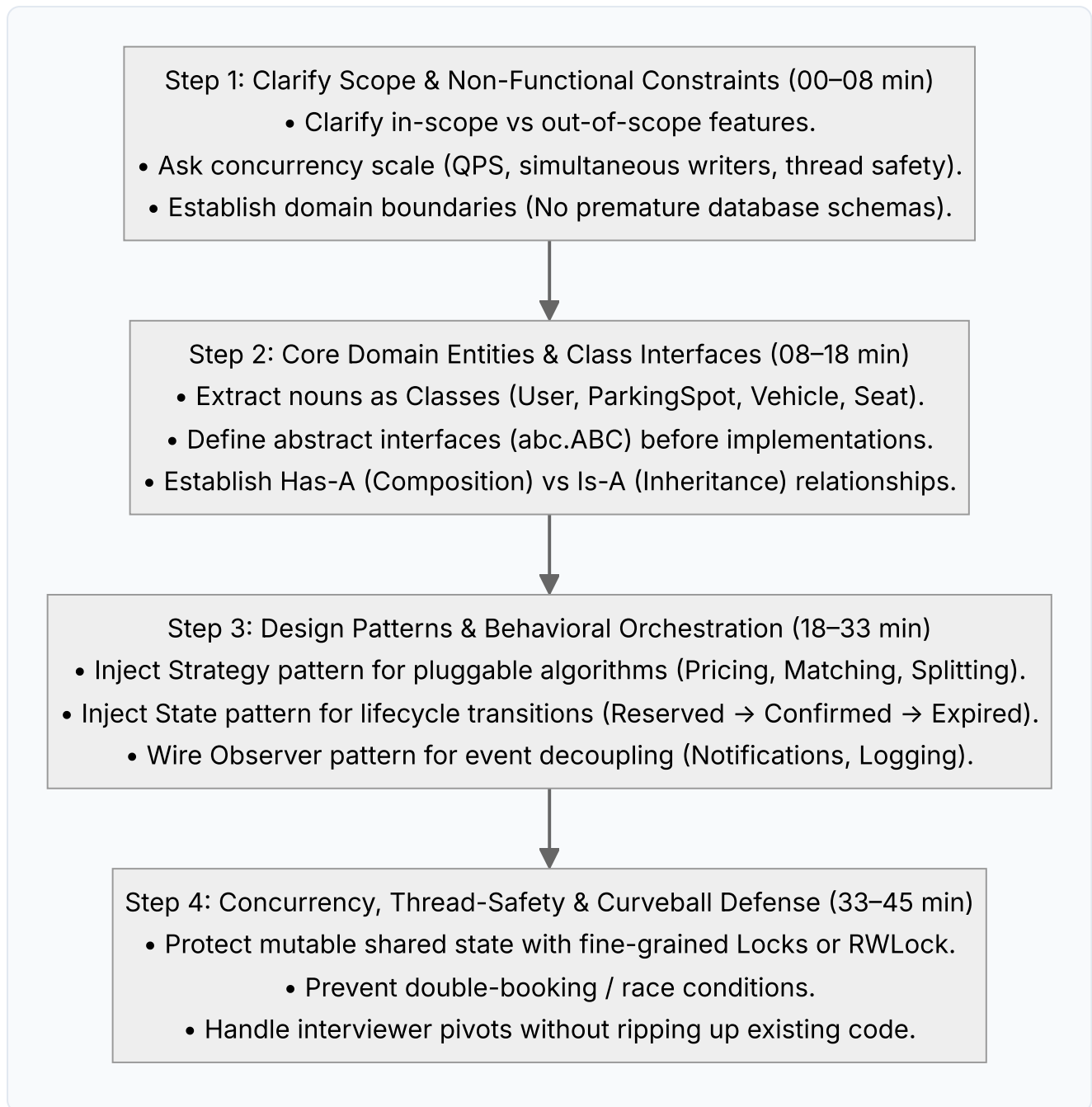


Chapter 13: Track 2 Recap — The LLD & Design Patterns Synthesis Playbook

Core Learning Objective: Consolidate everything you have mastered across Track 2 into a bulletproof, rapid-recall mental blueprint. This chapter provides a high-yield synthesis of the 4-step LLD interview framework, the 23 GoF and Concurrency patterns, a master comparison matrix across all 18 industrial systems, and the ultimate thread-safety synchronization cheat sheet.

1. The 4-Step LLD Interview Execution Engine

In a 45-minute Product-Based Company (Google, Meta, Uber, Amazon) LLD round, candidates fail when they start writing random class code immediately. Staff Engineers follow a structured, predictable 4-step execution rhythm:



2. The Master Pattern Decision Matrix

When faced with an object-oriented design problem, use this rapid pattern-matching table:

Category	Pattern	Plain-English Metaphor	When to Use in LLD Interviews	Core Systems Using It
Creational	Factory Method	Vending Machine Slot: you punch a code, it dispenses the correct snack without you knowing how it was wrapped.	Creating objects based on runtime inputs (Vehicle type, Notification type) without hardcoding constructors.	Parking Lot, Notification Engine
Creational	Builder	Custom Sub Sandwich: adding bread, cheese, toppings step-by-step until the sub is complete.	Constructing complex objects with 6+ optional parameters or immutability requirements.	Job Scheduler, Flight Reservation
Creational	Singleton	The Town Hall Clock: only one physical clock tower exists in the city; everyone reads the exact same time.	Managing single shared resources (DB Connection Pool, Configuration Registry).	Logger Framework, Lock Manager
Structural	Strategy	Interchangeable Phone Camera Lenses: snapping on a Macro, Fisheye, or Telephoto lens onto the same camera body.	Dynamically swapping algorithms at runtime (Pricing, Route Finding, Debt Splitting, Eviction policies).	Splitwise, Uber, Parking Lot, Rate Limiter
Structural	Observer	Subscribing to a YouTube Channel: when a new video uploads, all subscribers get a notification automatically.	Decoupling state changes from downstream side-effects (Email alerts, analytics, inventory deduction).	Pub/Sub, Inventory, Job Scheduler
Structural	State	Traffic Light: the light switches Green → Yellow → Red; button presses behave differently depending on current color.	Eliminating giant <code>if/elif/else</code> ladders for entities with lifecycles (Booking: Draft → Held → Paid → Cancelled).	Movie Booking, Vending Machine, Elevator
Structural	Decorator	Clothing Layers: putting on a t-shirt, adding a hoodie, adding a	Wrapping additional behaviors around objects dynamically	High-Throughput Logger

Category	Pattern	Plain-English Metaphor	When to Use in LLD Interviews	Core Systems Using It
		raincoat; you are still the same person, but now waterproof.	(Logging, Encryption, Compression, Toll-road fees).	
Structural	Composite	Russian Nesting Dolls / File Folders: a folder contains individual files AND subfolders, but both can report their size.	Tree hierarchies where individual items and groups of items share the same interface.	In-Memory File System
Concurrency	Readers-Writer Lock	Public Library Notice Board: 100 people can read the board at once, but when the librarian writes, everyone must wait.	High-read, low-write data structures (Seat map, File directory, Product catalog) achieving 15x throughput speedup.	File System, Seat Reservation
Concurrency	Producer-Consumer	Bakery Counter: bakers bake bread and put it on the shelf; customers take bread off the shelf; nobody waits directly on each other.	Decoupling fast requests from slow I/O workers using a bounded, thread-safe queue.	Logging Framework, Kafka Lite

3. The 18 Industrial Systems Comparison Matrix

Every system you mastered in Track 2 solves a distinct engineering challenge:

System #	System Name	Primary Patterns	Concurrency & Thread-Safety	Key Invariant / Algorithmic Trick
01	Distributed Job Scheduler	Strategy, Command, Observer	ThreadPoolExecutor + PriorityQueue	Min-heap ordered by next execution timestamp ($O(\log N)$).
02	Library Management	Observer, State, Factory	Re-entrant Lock (<code>threading.RLock</code>)	Multi-book checkout atomic reservations with due-date alerts.
03	Movie Booking System	State Machine, Strategy, Factory	Fine-grained Seat Mutex Locks	10-minute temporary seat hold with automatic timer expiry.
04	Car Rental System	Strategy, State, Factory	Pessimistic Reservation Locking	Inventory matching based on category and pickup/dropoff dates.
05	Parking Lot System	Factory, Strategy, Min-Heap	Per-Spot Re-entrant Locks	$O(\log N)$ nearest-spot allocation using Min-Heap per vehicle type.
06	Inventory Management	Observer, Strategy, State	Atomic Counter CAS / Locks	Low-stock threshold triggers auto-replenishment events.
07	Ride-Sharing (Uber LLD)	Strategy, Observer, State	Concurrent Driver Location Map	Haversine distance spatial matching + dynamic surge pricing engine.
08	API Rate Limiter	Strategy, Sliding Window	Sliding Window Log in deque	Token Bucket vs Sliding Window Log boundary burst defense.
09	Snake and Ladders	Factory, Strategy, State	Turn-based state coordination	Modulo board indexing and collision jumping.
10	Elevator System	State, Strategy, Command	SCAN algorithm dispatch queue	Minimizing total elevator travel distance using SCAN/LOOK queues.
11	Vending Machine	State Pattern Transition Engine	Coin/Note escrow synchronization	Physical item dispensing only when exact balance invariant holds.

System #	System Name	Primary Patterns	Concurrency & Thread-Safety	Key Invariant / Algorithmic Trick
12	In-Memory File System	Composite, Trie, Decorator	Custom <code>RWLock</code> on directory tree	Multi-reader concurrent file reads with exclusive writer locks.
13	High-Throughput Logger	Producer-Consumer, Strategy	Bounded Blocking Queue	Non-blocking async log append with background batch flushing.
14	Pub/Sub Message Broker	Observer, Strategy, Partitioning	Per-Partition Mutex Locks	Sequential append-only commit log with consumer group offsets.
15	Distributed Lock Manager	Redlock, Quorum, Fencing	Redis Quorum + Monotonic Tokens	Monotonic fencing tokens protecting against long GC pauses.
16	Kafka Consumer Rebalance	Sticky Assignor, State	Heartbeat timer thread coordination	Cooperative incremental rebalance avoiding stop-the-world pauses.
17	Splitwise Expense Sharing	Strategy, Greedy Min-Heap	User balance ledger locks	$O(N \log N)$ Min-Cash-Flow debt simplification via paired heaps.
18	Notification & Alerting	Adapter, PriorityQueue, Filter	PriorityQueue + Sliding Window Limiter	Multi-channel dispatch (Email/SMS/Push) + user fatigue rate limits.

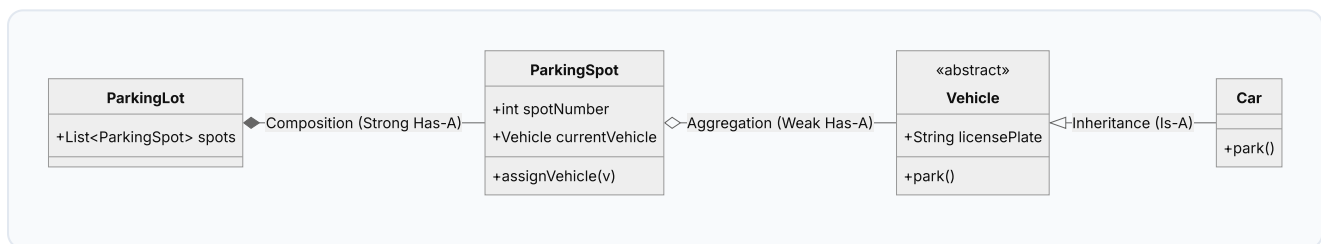
4. The Junior vs. Senior Antipattern Graveyard

#	The Junior Antipattern	What Goes Wrong in Production	The Senior / Staff Solution
1	The "God-Class" Monolith (e.g. <code>CinemaManager</code> handling seating, payments, emails, discounts, refunds in one 1,500-line class).	Violates Single Responsibility Principle (SRP) . Any change breaks unrelated features; impossible to unit test.	Decompose into distinct services: <code>SeatInventory</code> , <code>PricingEngine</code> , <code>PaymentProcessor</code> , and <code>NotificationDispatcher</code> .
2	Hardcoding <code>if vehicle_type == "CAR": fee = hours * 10</code> inside the business method.	Violates Open/Closed Principle (OCP) . Adding Electric Cars or Trucks requires modifying tested, deployed core classes.	Inject a <code>PricingStrategy</code> interface: <code>pricing_strategy.calculate_fee(duration)</code> .
3	Locking the entire <code>ParkingLot</code> or <code>CinemaTheater</code> with a single global Mutex.	Total serialization bottleneck: 10,000 users trying to view different movie seats or park in different floors block each other!	Lock only the specific entity (<code>Seat</code> or <code>ParkingSpot</code>), or use a <code>Readers-Writer Lock (RWLock)</code> for concurrent reads.
4	Using integer status flags (<code>status = 0</code> , <code>status = 1</code> , <code>status = 2</code>) scattered across <code>if/elif</code> statements.	State transitions become chaotic. Invalid states (e.g. paying for an already cancelled seat) slip through.	Use the State Pattern or an explicit finite State Machine enum where each state defines valid transition actions.

#	The Junior Antipattern	What Goes Wrong in Production	The Senior / Staff Solution
5	Acquiring multiple locks in arbitrary order across threads (Thread A: Lock1 → Lock2 , Thread B: Lock2 → Lock1).	Deadlock: both threads freeze permanently waiting for each other, requiring service restart.	Enforce a strict Global Lock Ordering convention (e.g. always acquire locks sorted by account ID).

5. The 5-Minute UML Drawing Survival Guide

When an interviewer asks you to sketch class relationships, use this exact mental cheat sheet:



- Inheritance (Is-A)**: **Car** is a **Vehicle** . Solid line with closed triangle arrow ($\triangleleft$ --).
- Composition (Strong Has-A)**: **ParkingLot** has **ParkingSpot** s. If the parking lot is bulldozed, the spots cease to exist. Filled diamond (* --).
- Aggregation (Weak Has-A)**: **ParkingSpot** has a **Vehicle** . If the spot is vacated or deleted, the vehicle still exists elsewhere. Open diamond (o --).
- Association / Dependency (Uses-A)**: A class calls a method on another class. Open arrow (— →).

6. Track 2 Graduation Milestone Check

Before you proceed to **Track 3: High-Level Design (HLD)**, confirm that you have mastered these 4 core competencies:

- ☒ Can you translate ambiguous product requirements into clean UML class diagrams and typed Python interfaces in under 15 minutes?

- ☒ Can you instantly identify whether a problem requires Strategy, Observer, State, Factory, or Composite?
- ☒ Can you explain why a Readers-Writer Lock provides a 15x throughput boost over a standard Mutex under read contention?
- ☒ Can you implement debt simplification (Min-Cash-Flow) or temporary seat reservation holds with zero race conditions?

TIP

Next Stop: Track 3 (High-Level Design)! You now know how to design rock-solid object-oriented code running inside a single machine or microservice. Now it is time to scale out to millions of users across globally distributed datacenters! Proceed to [Track 3: High-Level Design](#)!

Further Reading

- [Refactoring Guru: design patterns](#)
 - [Wikipedia: Design Patterns](#)
 - [UML diagrams reference](#)
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Check Yourself

Answer in your head or on paper first, then open each answer.

- ▶ 1. Which principle do most patterns serve?
- ▶ 2. What two diagrams do you draw in nearly every LLD interview?
- ▶ 3. Name three patterns that most often appear in LLD problems.
- ▶ 4. What is the most common mistake candidates make?